An Affirmative Answer to the Connected $C[0,1]$ BPBp-RSE Convexity Question

Automated Math Discovery Smoke Test

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Source and Open Question

This packet concerns Question 5.8 from:

> Helena del Rio, *Bollobas-type theorems for range strongly exposing operators*, arXiv:2512.10442.

The paper proves the corresponding necessity when the compact space is not connected, using non-trivial M-summands of $C(K)$. It then asks whether the same conclusion holds for the connected space $[0,1]$. The source location is page 31 of the paper.

> *with function* η_Y.
>
> We conclude the paper with some more open questions about the topic of the paper. The first one is naturally motivated by the results in Sections 3.2 and 3.3. Indeed, when K is connected, $C(K)$ does not have any M-summands and therefore Theorems 3.4 and 3.13 cannot be applied to the pair $(C(K), Y)$.
>
> **Question 5.8.** If the pair $(C[0,1], Y)$ has the BPBp-RSE in the complex case, does this imply that Y is $\mathbb{C}$-uniformly convex? In the real case, does this imply that Y is uniformly convex?

Quoted Tool from the Source Paper

We use the following lemma quoted in the source paper as Lemma 2.6.

Lemma 1 (RSE zero lemma)**.** *Let X and Y be Banach spaces. If $S \in \mathrm{RSE}(X,Y)$ attains its norm at $x_0 \in S_X$ and $z \in X$ satisfies*

$$\|x_0 + z\| \le 1 \quad \textit{and} \quad \|x_0 - z\| \le 1,$$

then $S(z) = 0$.

The Bump Functions

Let $X = C[0,1]$ over either $\mathbb{R}$ or $\mathbb{C}$. Choose $f, g \in X$ such that

$$0 \le f, g \le 1, \qquad \|f\| = \|g\| = 1, \qquad f(0) = 1, \qquad g(1) = 1,$$

and $\operatorname{supp} f \cap \operatorname{supp} g = \emptyset$. For instance, one may take two small triangular bump functions supported near 0 and 1. The only feature we use is this: if $x \in S_X$ and $\|x - f\| < \alpha < 1$, then there exists $\lambda > 0$ such that

$$\|x + \lambda g\| \leq 1 \quad \text{and} \quad \|x - \lambda g\| \leq 1.$$

Indeed, on $\operatorname{supp} g$ we have $f = 0$, hence $|x| < \alpha$ there, while outside $\operatorname{supp} g$ the perturbation by λg vanishes. Taking $0 < \lambda < 1 - \alpha$ gives the claim.

Main Result

Theorem 1. *Let Y be a Banach space.*

1. *In the complex case, if $(C[0,1], Y)$ has the BPBp-RSE, then Y is $\mathbb{C}$-uniformly convex.*
2. *In the real case, if $(C[0,1], Y)$ has the BPBp-RSE, then Y is uniformly convex.*

Consequently, combined with the sufficiency results in the source paper, the expected characterizations for the connected space $C[0,1]$ are valid.

Proof. We write η for a BPBp-RSE modulus of the pair $(C[0,1], Y)$.
Complex case. Fix $0 < \varepsilon < 1$ and put $\alpha = \varepsilon/3$. Choose

$$0 < \delta < \min\{\eta(\alpha), \alpha\}.$$

Let $y_0 \in S_Y$ and $y \in Y$ satisfy

$$\sup_{\omega \in \mathbb{T}} \|y_0 + \omega y\| \leq 1 + \delta.$$

We will prove that $\|y\| < \varepsilon$.

Define $T \colon C[0,1] \to Y$ by

$$T(h) = h(0) y_0 + h(1) y.$$

The norm of the associated map $\ell_\infty^2 \to Y$, $(a, b) \mapsto a y_0 + b y$, is attained on the distinguished boundary of the bidisc. Equivalently, the closed bidisc is the closed convex hull of $\mathbb{T}^2$, and the norm is convex. Therefore

$$\|T\| \leq \sup_{\substack{|a| \leq 1 \\ |b| \leq 1}} \|a y_0 + b y\| = \sup_{\omega \in \mathbb{T}} \|y_0 + \omega y\| \leq 1 + \delta.$$

Also $\|T\| \geq \|Tf\| = 1$. Hence

$$\left\| \frac{T}{\|T\|} f \right\| = \frac{1}{\|T\|} \geq \frac{1}{1 + \delta} > 1 - \eta(\alpha).$$

Applying the BPBp-RSE to $T / \|T\|$ and f, we obtain $S \in \mathrm{RSE}(C[0,1], Y)$ and $x \in S_{C[0,1]}$ such that

$$\|Sx\| = 1 = \|S\|, \qquad \left\| S - \frac{T}{\|T\|} \right\| < \alpha, \qquad \|x - f\| < \alpha.$$

By the bump observation, choose $\lambda > 0$ with $\|x \pm \lambda g\| \leq 1$. The RSE zero lemma gives $S(\lambda g) = 0$, and hence $Sg = 0$. Consequently,

$$\|y\| = \|Tg\| = \|Tg - Sg\| \leq \|T - S\| \leq \left\| S - \frac{T}{\|T\|} \right\| + \left\| \frac{T}{\|T\|} - T \right\| < \alpha + \delta < \varepsilon.$$

This is precisely the $\mathbb{C}$-uniform convexity estimate.

Real case. Fix $0 < \varepsilon < 1$ and put $\alpha = \varepsilon/4$. Choose

$$0 < \delta < \min\{\eta(\alpha), \alpha\}.$$

Let $u, v \in S_Y$ satisfy

$$\left\| \frac{u+v}{2} \right\| > 1 - \delta.$$

Set

$$a = \frac{u+v}{2}, \qquad b = \frac{u-v}{2},$$

and define $T \colon C[0,1] \to Y$ by

$$T(h) = h(0)a + h(1)b.$$

For $h \in B_{C[0,1]}$, the real numbers $r = h(0)$ and $s = h(1)$ satisfy $|r|, |s| \leq 1$, and

$$T(h) = \frac{r+s}{2} u + \frac{r-s}{2} v.$$

Since

$$\left| \frac{r+s}{2} \right| + \left| \frac{r-s}{2} \right| = \max\{|r|, |s|\} \leq 1,$$

we get $\|T\| \leq 1$. On the other hand, $\|T\| \geq \|Tf\| = \|a\| > 1 - \delta$. Thus

$$\left\| \frac{T}{\|T\|} f \right\| = \frac{\|a\|}{\|T\|} > 1 - \eta(\alpha).$$

Applying the BPBp-RSE to $T/\left\|T\right\|$ and f, we obtain $S \in \mathrm{RSE}(C[0,1], Y)$ and $x \in S_{C[0,1]}$ such that

$$\|Sx\| = 1 = \|S\|, \qquad \left\| S - \frac{T}{\|T\|} \right\| < \alpha, \qquad \|x - f\| < \alpha.$$

As above, the bump observation and the RSE zero lemma imply $Sg = 0$. Hence

$$\|b\| = \|Tg - Sg\| \leq \|T - S\| \leq \left\| S - \frac{T}{\|T\|} \right\| + \left\| \frac{T}{\|T\|} - T \right\| < \alpha + \delta < 2\alpha.$$

Therefore

$$\|u - v\| = 2\left\|b\right\| < 4\alpha = \varepsilon.$$

This proves uniform convexity of Y. □

Sanity Check Script

The file `code/check_bump_test.py` performs finite-dimensional numerical checks of the two elementary norm estimates used above. It does not constitute part of the proof; it is only a guard against sign or coefficient mistakes in the two-coordinate test operators.